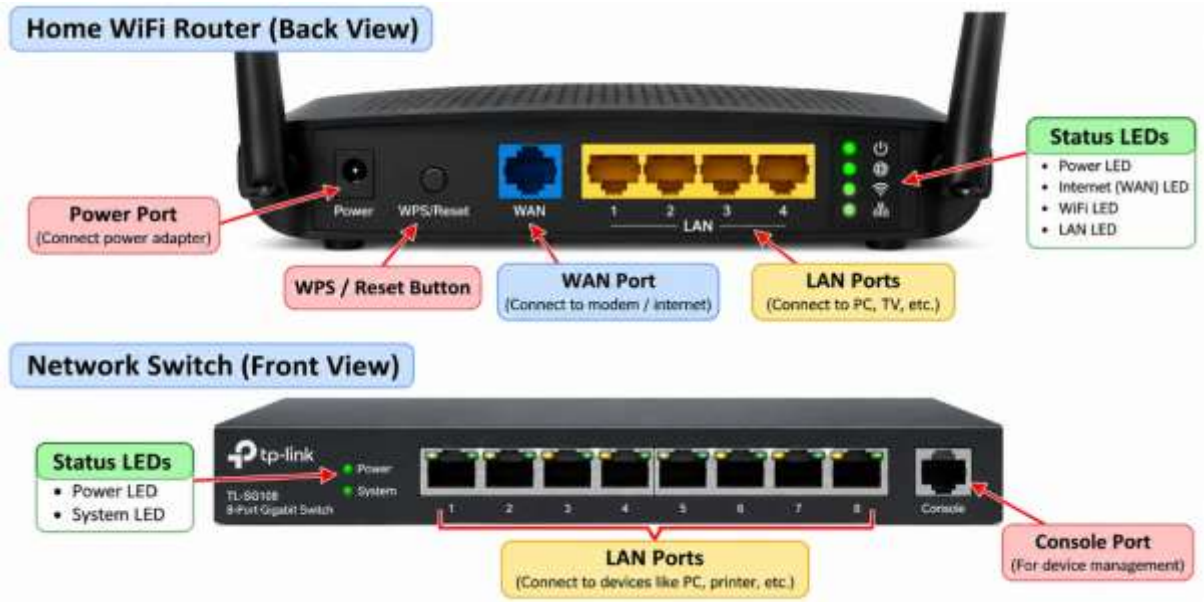
Day 2: Routers and Switches Theory (30 min): Basic functions and differences between routers and switches.Practical (30 min):Identify ports and LEDs on a router and switch.

1.Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

ANS



**2.List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.**

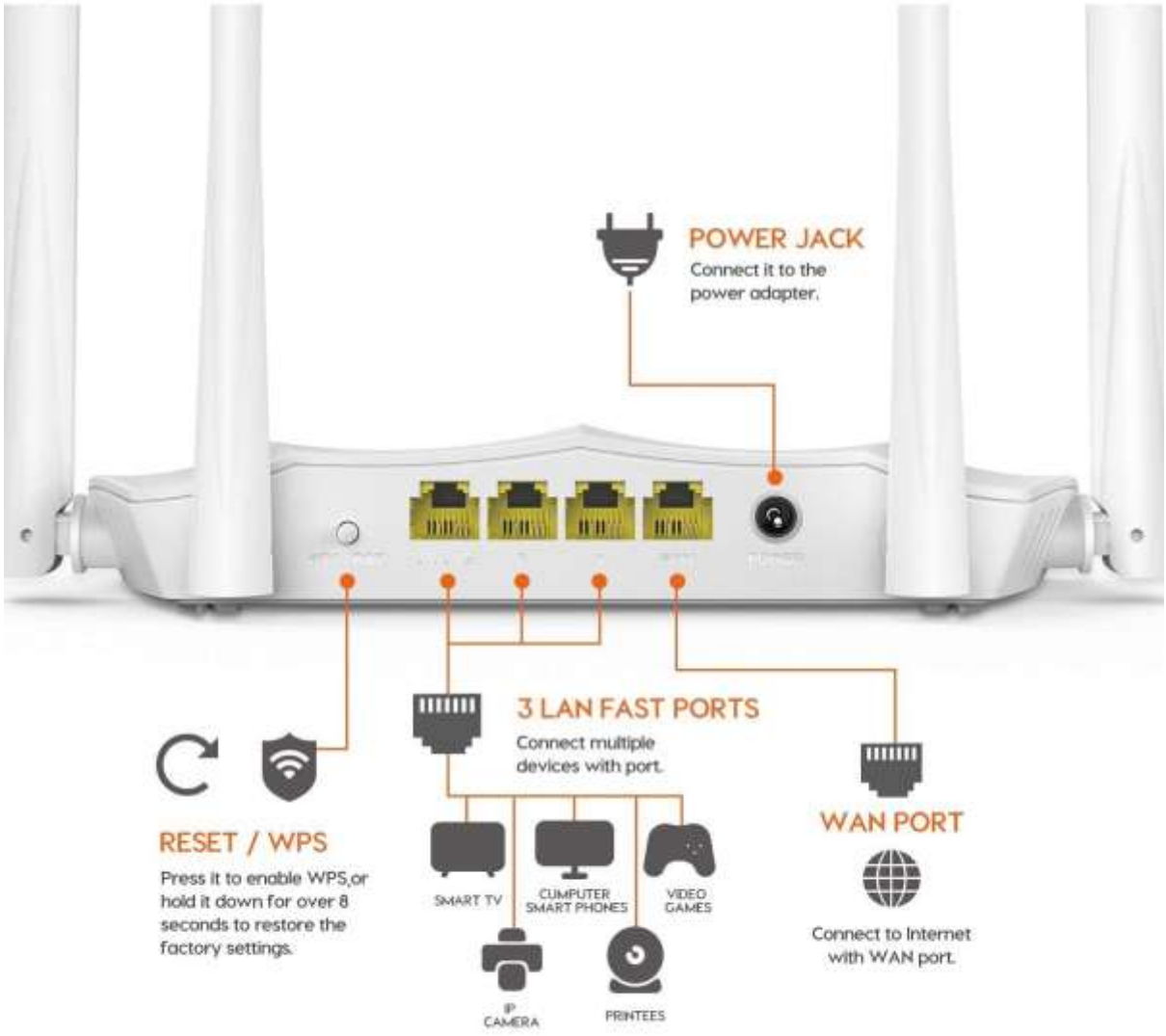
ANS:

Router	**Switch**
1. Connects different networks.	1. Connects computers and other devices.
2. Uses IP addresses to send data.	2. Uses MAC addresses to send data.
3. Connects a home network to the Internet.	3. Connects devices like computers and printers in one network.

3.Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

ANS:

1. **Router**



Identification and purpose:

1. Power port: Usually at the back of the router. It connects the power adapter to turn on the router.
2. Ethernet ports: Usually at the back, marked LAN 1, LAN 2, LAN 3, and LAN 4. They connect computers and other devices using Ethernet cables.

3. LED indicators: Usually on the front or top. They show the power status, internet connection, Wi-Fi status, and network activity.

2. Switch



Identification and purpose:

1. Power port: Usually at the back or side of the switch. It connects the power adapter to turn on the device.
2. Ethernet ports: Usually on the front or back, with multiple ports. They connect computers, printers, and other network devices.
3. LED indicators: Usually above or near the Ethernet ports. They show whether a device is connected and whether data is being transferred.

4.Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

ANS: Setting Up a Gaming Café Network

First, I will connect the internet line to the router. Then, I will connect the router to a switch using a cable. After that, I will connect all 8 gaming PCs to the switch using Ethernet cables.

Reason: The router provides internet, and the switch connects all 8 computers. This setup helps us play games and watch IPL matches smoothly.